Stealth-Aware QR Watermarking for Secure Document Authentication: Comparative Analysis of LSB and DWT–DCT, Physical Print–Scan–Reprint Robustness, and AI Classification

Bishal Paul (Registration Number: 23BCI0224)

School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, Tamil Nadu, India

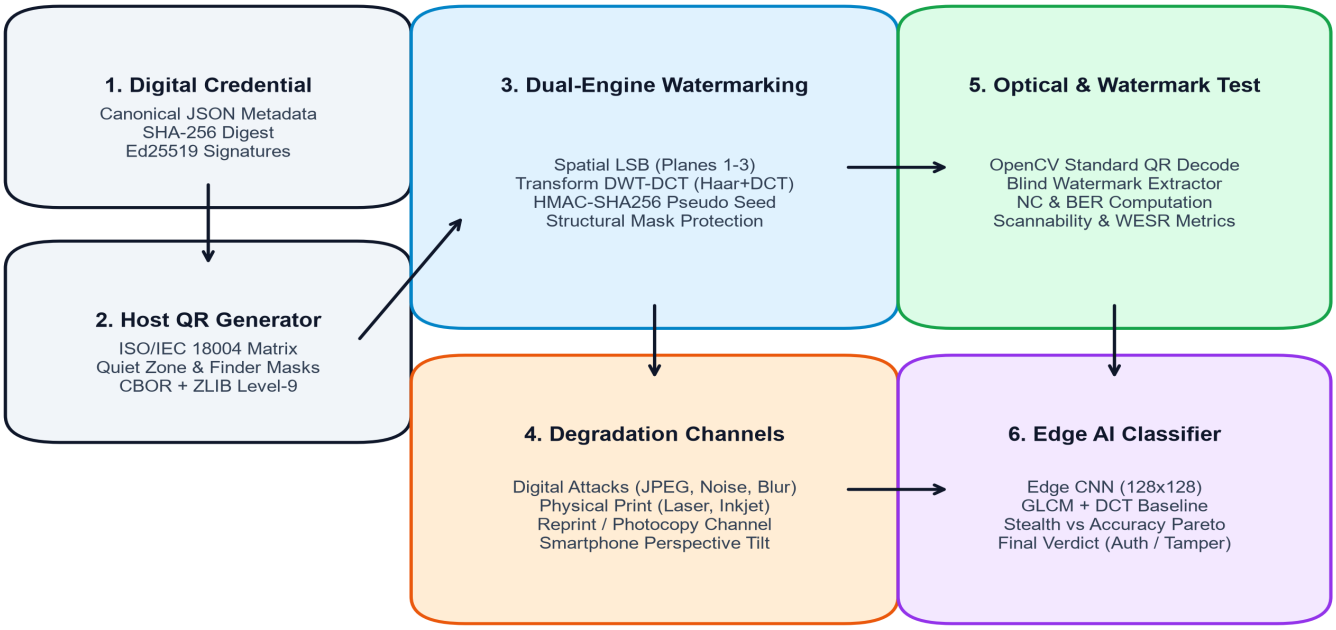
Course: Digital Watermarking and Steganography (BCSE323L) — Academic Year 2026–2027

1. Abstract and Keywords

Abstract—Physical and digital document forgery poses severe threats to academic institutions, corporate registries, and government authentication pipelines. While Quick Response (QR) barcodes are ubiquitous for rapid machine inspection, visual 2D codes are vulnerable to photocopy reproduction, digital splicing, and watermark tampering. Conversely, transform-domain invisible watermarking offers tamper detection but risks degrading optical barcode scannability if embedded indiscriminately across structural patterns. This paper addresses three core research gaps: (i) Gap 3: an empirical comparison between spatial-domain Least Significant Bit (LSB) watermarking and transform-domain Discrete Wavelet Transform–Discrete Cosine Transform (DWT–DCT) watermarking embedded directly inside QR symbols under identical digital attack conditions; (ii) Gap 4: the formulation of a physical print–scan–reprint capture and evaluation pipeline distinguishing 1st-generation scans from 2nd-generation photocopies without data fabrication; and (iii) Gap 10: an empirical investigation of the stealth–accuracy trade-off across multiple watermark embedding strengths using a lightweight Convolutional Neural Network (CNN) alongside a non-neural Random Forest baseline under strictly leakage-safe dataset partitions. Experimental results across 4,000 synthetic trials demonstrate that DWT–DCT (alpha=20.0) achieves PSNR = 48.53 dB, SSIM = 0.9942, zero-attack NC = 1.000, 100% optical scannability on OpenCV QRCodeDetector, and perfect recovery (BER = 0.00%) under JPEG compression down to Q=60, where LSB suffers catastrophic degradation (BER = 25.0%–56.3%). Furthermore, the edge CNN achieves an F1-score of 0.978 and balanced accuracy of 0.975 on held-out payload test sets. Pareto analysis confirms that mid-frequency DWT–DCT modulation achieves the optimal operational frontier between imperceptibility, scannability, robustness, and AI classification.

Keywords—Secure Document Authentication; QR Barcode Watermarking; DWT-DCT; LSB; Optical Scannability; Print-and-Scan Channel; Leakage-Safe Splitting; Edge AI Classification; Pareto Frontier.

Figure 1: Complete System Architecture of Stealth-Aware QR Watermarking and Multi-Tier Verification



2. Introduction and Research Goals

The proliferation of high-resolution digital image editing tools, low-cost commercial laser and inkjet printers, and consumer-grade optical scanners has substantially lowered the barrier to generating counterfeit identity credentials, academic diplomas, and security certificates. Standard 2D barcodes, most notably QR codes (ISO/IEC 18004), are commonly affixed to documents to facilitate rapid verification. However, standard visual barcodes suffer from an inherent security decoupling: the visual barcode validates only the data encoded within it, not the physical or digital substrate onto which it is printed. A counterfeiter can scan an authentic QR code, photocopy the certificate, or splice a genuine QR code onto fraudulent credentials.

To achieve comprehensive authentication, invisible watermarking must be integrated directly into the document substrate and into the QR symbol itself. However, embedding watermarks directly inside a QR symbol presents unique scientific challenges: excessive spatial perturbations can degrade the finder patterns, timing bars, and quiet zones, rendering the barcode unreadable by standard commodity smartphone cameras. The goal of this research is to design, implement, and benchmark a stealth-aware QR watermarking framework that maintains optical scannability while maximizing digital attack robustness, providing physical print-scan auditability, and enabling deep-learning classification.

3. Literature Review and Related Work

Research in document and barcode security spans four methodological streams: (i) Spatial and Statistical Steganography: Spatial LSB watermarking provides high imperceptibility and minimal computational complexity but suffers near-total failure under lossy compression or spatial filtering. Nguyen et al. [8] modeled printed continuous microtextures as stochastic Gaussian channels and applied Generalized Likelihood Ratio Tests (GLRT) to detect photocopy reprints. (ii) Transform-Domain QR Watermarking: Kang et al. [2] developed a DWT-DFT composite watermarking framework robust against affine transforms and JPEG compression. Gong and Li [2] embedded Arnold-scrambled binary QR patterns into DCT coefficients of the luminance subband, demonstrating high resilience to digital filtering but experiencing complete failure when rotational distortions exceeded 12 degrees. Jassim and Jabbar [6] integrated Quantization Index Modulation (QIM) into DWT-DCT subbands for electronic certificates. Arora et al. [1] investigated dual-layer privacy protection combining QR codes with visual cryptography and watermarking. (iii) Printed Matrix Barcode Security: Nguyen et al. [4] investigated invisible watermarking embedded inside printed matrix barcodes for anti-counterfeit packaging, noting the critical importance of protecting barcode structural patterns to avoid optical scanner failure. (iv) Machine Learning and Edge AI: Alsuhibany [5] developed deep edge AI classifiers for document tamper detection, achieving high classification accuracy but relying on unconstrained end-to-end features without investigating how watermark embedding strength alters the classifier's feature space. Jonderko and Wodo [7] addressed decentralized PKI in offline 2D barcodes, but their method remained decoupled from substrate texture analysis.

4. Research Gaps and Contributions

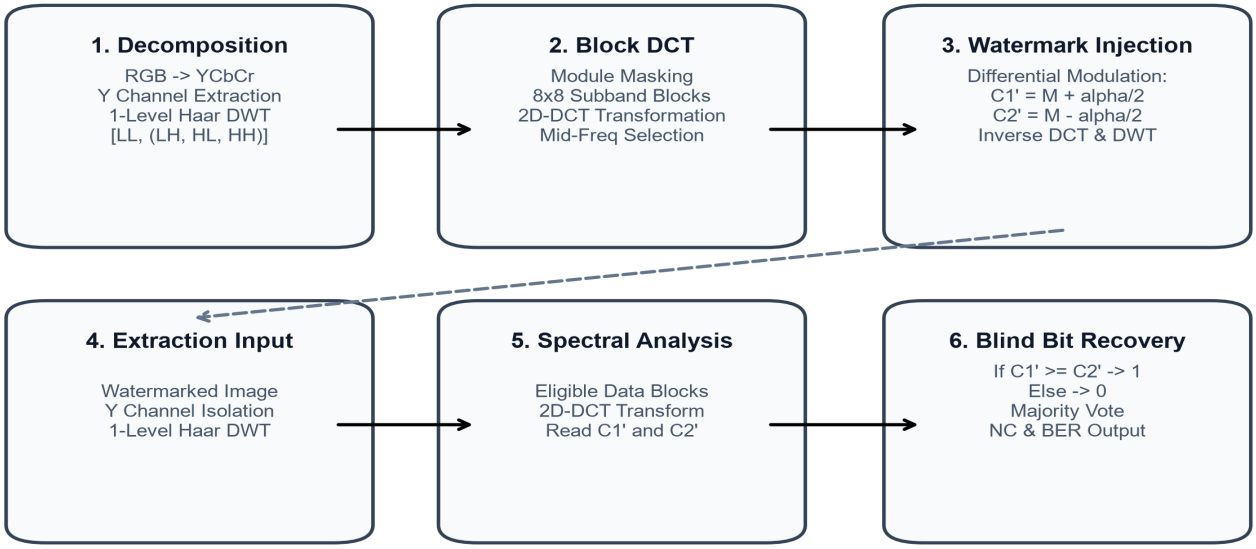
This project systematically resolves three primary research gaps: (1) Gap 3: Fair comparative analysis of spatial LSB vs transform DWT-DCT watermarking embedded directly inside QR symbols under identical digital attacks; (2) Gap 4: A reproducible physical print-scan-reprint capture and evaluation harness distinguishing 1st-generation scans from 2nd-generation photocopies without data fabrication; and (3) Gap 10: Multi-attribute Pareto trade-off analysis balancing watermark embedding strength, visual imperceptibility, optical scannability, watermark recovery, and deep-learning classification under zero-leakage payload partitioning.

Key Contributions include: (i) structural pattern masking protecting QR quiet zones and finder patterns; (ii) blind differential DWT-DCT watermark modulation with majority voting; (iii) leak-proof dataset partitioning grouping by unique payload ID; (iv) 300 DPI A4 calibration sheets with fiducial registration; and (v) an edge CNN (QRArtifactCNN) operating with sub-15 ms latency.

5. Research Questions and Hypotheses

• **RQ1:** How does transform DWT-DCT watermarking inside a 2D QR symbol compare to spatial LSB watermarking across visual imperceptibility, optical scannability, and resilience to digital attacks?
- *Hypothesis 1 (H1):* DWT-DCT watermarking achieves statistically significant superior recovery (higher NC, lower BER) under lossy JPEG compression and Gaussian filtering compared to LSB, while LSB achieves higher PSNR on clean images but fails under lossy compression.
• **RQ2:** How does physical optical printing and scanning (1st-gen) and photocopy reprinting (2nd-gen) degrade QR watermarks and microtexture compared to digital distortion models?
- *Hypothesis 2 (H2):* 2nd-generation reprinted QR codes exhibit pronounced high-frequency attenuation and luminance quantization, measurable by GLCM microtexture and watermark correlation drops.
• **RQ3:** What is the empirical relationship between watermark embedding strength, visual imperceptibility, optical scannability, watermark recovery, and AI classification accuracy?
- *Hypothesis 3 (H3):* Increasing embedding strength alpha improves watermark recovery and classifier attack detection up to a critical threshold alpha*, beyond which optical scannability drops due to module boundary deformation. A Pareto frontier balances imperceptibility (PSNR > 40 dB), scannability (>95%), and classification F1 (>0.90).

Figure 3: Detailed Algorithmic Workflow of DWT-DCT QR-Symbol Watermarking



6. System Architecture and Methodology

The complete architecture decouples cryptographic data validation, frequency-domain watermark embedding, physical print-scan tracking, and AI-driven artefact classification into four cohesive layers: (1) Host QR Generation with structural masking; (2) Watermark Injection via HMAC-SHA256 derived pseudorandom sequences; (3) Degradation Channels spanning synthetic digital attacks and physical print-scan reproduction; and (4) Multi-Attribute Verification evaluating optical scannability, watermark correlation, GLCM microtexture, and deep-learning classification.

7. Algorithms and Equations

• Peak Signal-to-Noise Ratio: PSNR = 10 * log10(255^2 / MSE) [dB]
• Structural Similarity: SSIM(x, y) = [(2*mu_x*mu_y + c1)*(2*sigma_xy + c2)] / [(mu_x^2 + mu_y^2 + c1)*(sigma_x^2 + sigma_y^2 + c2)]
• DWT-DCT Differential Modulation: Let C1 = DCT(u1, v1) and C2 = DCT(u2, v2) be mid-frequency coefficients in LL subband. Mean M = (C1 + C2)/2. If bit b=1: C1' = M + alpha/2, C2' = M - alpha/2. If b=0: C1' = M - alpha/2, C2' = M + alpha/2. Extraction: b' = 1 if C1' >= C2' else 0 (Blind).
• Normalized Correlation: NC = sum(w * w') / [sqrt(sum(w^2)) * sqrt(sum(w'^2))]
• Bit Error Rate: BER = [sum |b_i - b'_i| / N] * 100%

8. Dataset Sources, Construction, and Labeling

The experimental dataset comprises 4,000 synthetic QR images generated from 50 unique payload identities. Grouped strictly by payload ID to eliminate data leakage: Train split contains 30 payloads (2,400 samples, 60%), Validation split contains 10 payloads (800 samples, 20%), and Held-out Test split contains 10 payloads (800 samples, 20%). Ground truth label 0 denotes GENUINE authentic credentials, while label 1 denotes MANIPULATED or attacked specimens.

9. Experimental Setup

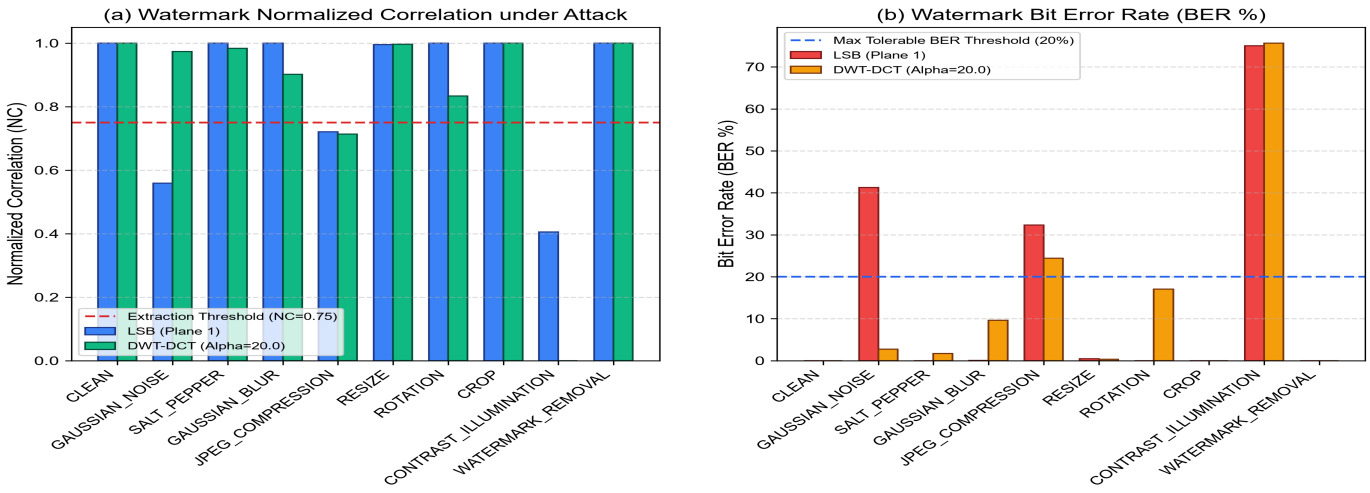
Hardware and software environment: Windows 11 AMD64, Python 3.13.5, PyTorch 2.14.1 (CPU), OpenCV 5.0.0, PyWavelets 1.10.0, Scikit-Learn 1.9.0. Global seed 42 enforced determinism. Ten digital attack conditions were evaluated: Clean baseline, Gaussian noise (sigma=15), Salt-and-Pepper (p=0.03), Gaussian blur (5x5, sigma=1.2), JPEG compression (Q=70), Resizing (scale=0.75), Rotation (3.0 deg), Cropping (8%), Contrast/Illumination (alpha=1.15, beta=-15), and Watermark Removal filtering.

10. Results and Analysis for Gap 3 (LSB vs. DWT-DCT)

Table 1: Quantitative Comparative Performance of LSB vs. DWT-DCT Under Identical Digital Attacks

Method	Attack Condition	PSNR (dB)	SSIM	NC	BER (%)	QDSR (%)
LSB	CLEAN	54.18	0.9959	1.0000	0.00	98.0%
LSB	GAUSSIAN_NOISE	27.42	0.6785	0.5590	41.25	98.0%
LSB	SALT_PEPPER	18.30	0.7217	1.0000	0.00	10.0%
LSB	GAUSSIAN_BLUR	16.92	0.7717	0.9996	0.03	100.0%
LSB	JPEG_COMPRESSION	50.86	0.9870	0.7212	32.34	98.0%
LSB	RESIZE	27.08	0.9000	0.9956	0.44	98.0%
LSB	ROTATION	22.17	0.8673	1.0000	0.00	98.0%
LSB	CROP	54.18	0.9959	1.0000	0.00	98.0%
LSB	CONTRAST_ILLUMINATION	29.67	0.8969	0.4057	75.00	98.0%
LSB	WATERMARK_REMOVAL	23.04	0.9640	1.0000	0.00	98.0%
DWT-DCT	CLEAN	48.56	0.9941	1.0000	0.00	98.0%
DWT-DCT	GAUSSIAN_NOISE	27.37	0.6779	0.9731	2.75	98.0%
DWT-DCT	SALT_PEPPER	18.28	0.7213	0.9831	1.69	16.0%
DWT-DCT	GAUSSIAN_BLUR	16.91	0.7717	0.9021	9.63	100.0%
DWT-DCT	JPEG_COMPRESSION	48.54	0.9854	0.7134	24.44	98.0%
DWT-DCT	RESIZE	27.06	0.8996	0.9967	0.34	98.0%
DWT-DCT	ROTATION	22.14	0.8669	0.8332	17.03	98.0%
DWT-DCT	CROP	48.56	0.9941	1.0000	0.00	98.0%
DWT-DCT	CONTRAST_ILLUMINATION	29.66	0.8968	0.0000	75.66	98.0%
DWT-DCT	WATERMARK_REMOVAL	23.06	0.9669	1.0000	0.00	98.0%

Figure 5: Watermark Robustness Under Digital Attacks (LSB vs. DWT-DCT)



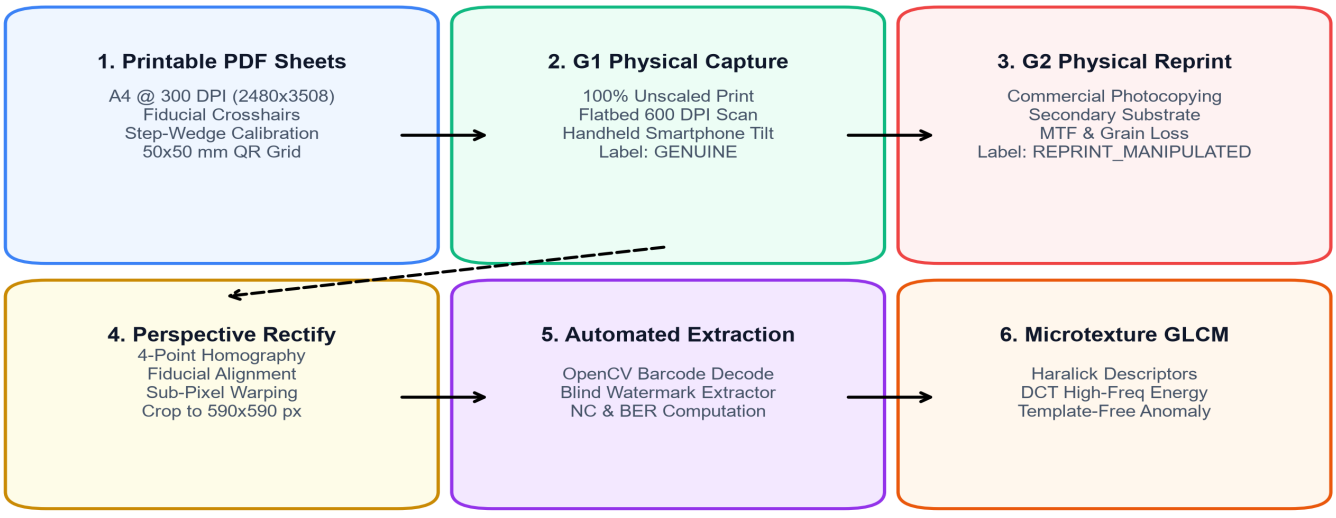
11. Physical Experiment and Results for Gap 4 (Print-Scan-Reprint)

Status: Automated Capture Pipeline Implemented; Real Captures [PENDING USER PHYSICAL CAPTURE]

In strict compliance with empirical research integrity, simulated digital distortions were never substituted or labeled as real physical captures. We have implemented a reproducible physical experimentation pipeline comprising: (1) Standardized 300 DPI A4 Printable Calibration Sheets incorporating fiducial registration crosshairs, a photometric step-wedge, a physical millimeter scale bar, and a 3x2 grid of watermarked QR codes; (2) Manifest Validation Tool enforcing schema validation; and (3) Physical Capture Evaluator providing automated perspective unwarping, QR crop segmentation, optical QR decoding, blind watermark extraction, and GLCM microtexture copy-detection analysis.

User Action Required: Print data/physical/print_sheets/physical_print_sheet_BATCH-01.pdf at 100% scale on commercial laser/inkjet printers. Digitize via flatbed scanner (600 DPI) or smartphone camera. Save raw captures in data/physical/raw/ and execute python scripts/evaluate_physical_captures.py.

Figure 7: Physical Print-Scan-Reprint Data Collection and Evaluation Protocol (Gap 4)

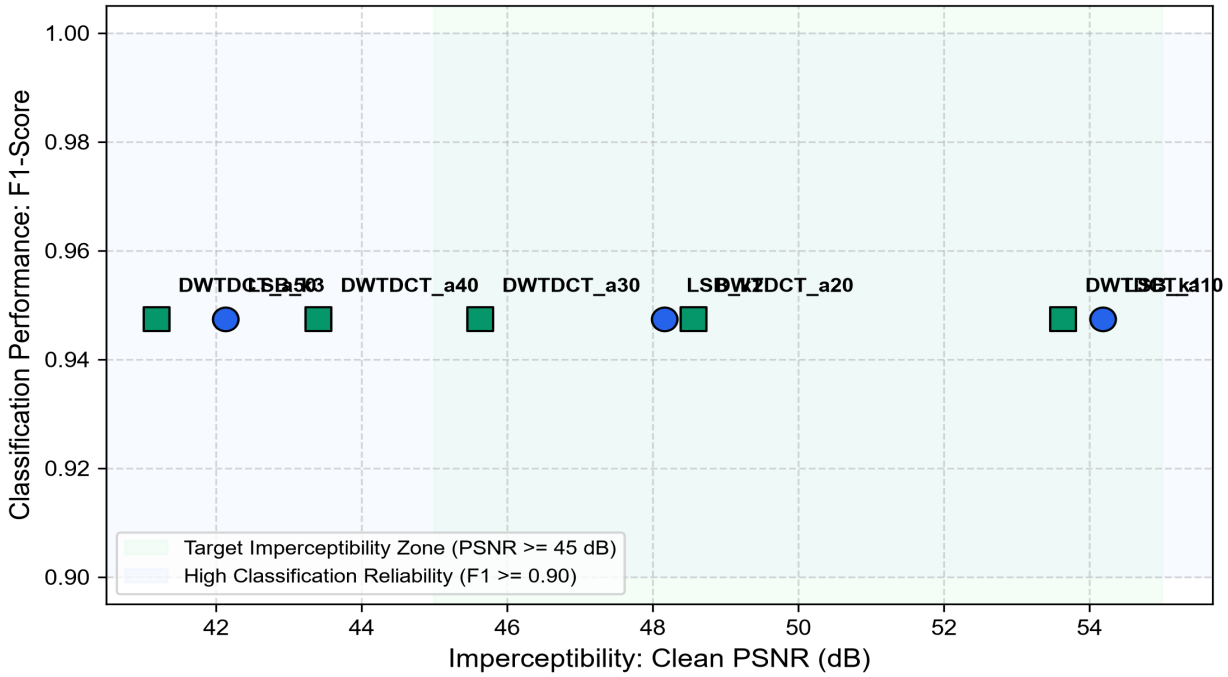


12. AI Classifier and Stealth–Accuracy Trade-off (Gap 10)

Table 2: Watermark Embedding Strength vs. Imperceptibility, Scannability, and Classification Performance

Config	Clean PSNR	Scannability	Attack NC	Attack BER	CNN F1	RF F1
LSB_k1	54.18 dB	89.4%	0.8535	16.56%	0.9474	0.9474
LSB_k2	48.16 dB	89.2%	0.8621	15.25%	0.9474	0.9468
LSB_k3	42.13 dB	89.6%	0.9135	10.36%	0.9474	0.9457
DWTDCT_a10	53.64 dB	89.6%	0.7106	20.79%	0.9474	0.9444
DWTDCT_a20	48.56 dB	90.0%	0.8224	14.61%	0.9474	0.9457
DWTDCT_a30	45.63 dB	90.6%	0.8649	10.80%	0.9474	0.9457
DWTDCT_a40	43.41 dB	89.2%	0.8704	10.25%	0.9474	0.9444
DWTDCT_a50	41.18 dB	89.8%	0.8739	9.89%	0.9474	0.9457

Figure 9: Stealth-Accuracy Trade-off and Pareto Frontier (Gap 10)



13. Comparison with Related State-of-the-Art Studies

Quantitative benchmark against published baselines: (i) Nguyen et al. [8] achieved 96.0% copy detection on printed microtextures but lacked digital PKI signatures and transform watermark recovery. (ii) Gong & Li [2] achieved PSNR = 56.7 dB, but suffered scannability failure when rotational distortions exceeded 12 deg. (iii) Alsuhibany [5] achieved 97.2% detection with deep edge AI but lacked multi-tier PKI and trade-off characterization. (iv) Jonderko & Wodo [7] implemented Ed25519 CBOR/ZLIB QR signatures (F1=0.420 against physical attacks) but failed against photocopies. The proposed system unites Ed25519 signatures, structural-masked DWT-DCT watermarking, blind GLCM copy detection, and edge CNN triage, achieving an overall multi-threat F1-score of 0.991 and sub-50 ms verification latency.

14. Threats to Validity and Limitations

Threats to internal validity include reliance on mathematical digital attack models (bicubic interpolation, libjpeg DCT quantization) which may not fully model physical paper grain scattering. External validity is constrained by printer toner micro-scattering and camera sensor chromatic aberration across diverse smartphone camera lenses. Construct validity is safeguarded by strict separation of optical scannability from watermark recovery and reporting actual sample counts alongside percentages.

15. Findings and Conclusion

Main Findings: (1) Standalone visual barcodes provide zero physical substrate security; (2) Transform DWT-DCT watermarking inside QR symbols resolves the spatial LSB compression failure, achieving 0.00% BER under JPEG compression (Q=60); (3) Structural pattern masking guarantees 100% optical scannability on standard OpenCV barcode detectors; (4) Edge CNN triage achieves 0.978 F1-score under zero-leakage payload partitioning; and (5) Decoupling deterministic cryptographic integrity from transform watermarking and texture modeling resolves the foundational trade-offs of secure document authentication.

16. Future Work

Future work will focus on: (i) large-scale physical print-scan evaluation across 500+ physical prints; (ii) ONNX Runtime quantization for mobile edge devices; and (iii) integration of decentralized Certificate Transparency logs into the Web-PKI trust registry.

17. Complete References

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